Several industries now run reinforcement learning in production, not just in simulation. I focus on systems where RL operates continuously in the field, affecting millions of real decisions, such as ride-hailing dispatch at DiDi, data center cooling at Google, hotel revenue management, and financial order execution. The chapter concludes with a simulation study on the bus engine replacement problem, demonstrating that RL matches and surpasses dynamic programming on a classical benchmark.

0.1 Ride-Hailing Dispatch

Each driver-passenger assignment changes the spatial distribution of available drivers, making ride-hailing dispatch a sequential optimization problem at a scale (tens of millions of daily rides at DiDi) where exact dynamic programming is intractable.

Qin et al. (2021) formalized DiDi's order dispatching as a semi-Markov decision process[1] where each driver is an independent agent. The state of a driver consists of location (discretized into hexagonal zones) and time (bucketed into intervals). The action is the order assigned to the driver, with the option to remain idle. The reward is the trip fare. State transitions are determined by trip destinations, as completing an order transports the driver from origin to destination, changing their spatial state. The stochasticity arises from future demand, which determines the available actions at each state. The per-driver value function $V^\pi(s)$ represents the expected cumulative fare a driver can earn from state s under dispatching policy π:

$$V^\pi(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^{\tau_k} r_k \mid s_0 = s, \pi\right], \tag{1}$$

where τ_k is the time to complete the k-th trip and r_k is the corresponding fare. The platform's objective is to maximize total driver income across the fleet, which decomposes into the sum of individual driver value functions.

The deployed system uses generalized policy iteration within a combinatorial optimization framework. Each dispatching window (a few seconds), the platform collects open orders and available drivers, constructs a bipartite graph, and solves a linear assignment problem. The edge weights are computed as the advantage of each driver-order

[1] A semi-Markov decision process generalizes the standard MDP by allowing variable time between decisions. The discount factor γ^{τ_k} depends on the actual time elapsed τ_k rather than applying a fixed per-step discount.

match relative to the driver's current state value:

$$w_{ij} = \hat{r}_i + \gamma^{\hat{\tau}_i} V(s'_j) - V(s_j), \tag{2}$$

where $\hat{r}_i$ is the predicted fare for order i, $\hat{\tau}_i$ is the estimated trip duration, and s'_j is the destination state. The value function is learned via temporal-difference methods from historical trip data. DiDi's Cerebellar Value Network (CVNet; Tang et al., 2019) uses hierarchical coarse-coding[2] with a multi-resolution hexagonal grid, enabling transfer learning across cities and robustness to data sparsity in low-traffic zones.

Production deployment across more than 20 Chinese cities demonstrated consistent improvements. Table 1 summarizes the reported gains from A/B tests using time-slice rotation, where algorithms alternate control of the platform in 3-hour blocks to avoid interference effects.

Table 1: DiDi dispatch deployment results from Qin et al. (2021) and Tang et al. (2019).

Metric	Improvement vs. Baseline	Scale
Total driver income	+0.5–2.0%	Tens of millions daily rides
Order response rate	+0.5–2.0%	Platform-wide
Fulfillment rate	+0.5–2.0%	Platform-wide

Li et al. (2019) addressed the coordination challenge using mean-field multi-agent RL. With thousands of drivers making simultaneous decisions, the full multi-agent state space is intractable. The mean-field approximation replaces individual driver states with an aggregate distribution, allowing each driver to condition on the density of nearby drivers rather than their exact locations. Experiments on DiDi data showed improved fleet utilization compared to single-agent baselines.

Han et al. (2022) reported complementary results from Lyft, where the dispatching system optimizes driver assignment and repositioning jointly. Their value decomposition architecture[3] assigns credit to individual driver decisions within the global objective. The production system demonstrated improvements in rider wait times and driver utilization, confirming that the RL dispatch paradigm generalizes across platforms and regulatory

[2]Coarse-coding represents a continuous state using overlapping receptive fields at multiple resolutions. At coarser levels, each tile covers a large region, capturing broad spatial patterns; finer levels capture local detail. The value function is the sum of features activated across all resolution levels.

[3]Value decomposition decomposes the platform's global matching objective into individual driver value functions, each estimable via temporal-difference learning. The global optimum is recovered by optimizing the sum of individual values.

environments.

The percentage improvements appear modest, but at the scale of ride-hailing operations they translate to substantial aggregate effects. A 1% improvement in fulfillment rate at DiDi's volume represents hundreds of thousands of additional completed rides per day. The central lesson from these deployments is that dispatching is not just a matching problem but a fleet positioning problem. Today's assignments determine tomorrow's driver distribution.

0.2 Data Center Cooling

Data centers consume enormous amounts of energy, with cooling systems accounting for 30–40% of total power usage. Google applied machine learning to autonomous HVAC control, achieving substantial efficiency gains while maintaining temperature safety constraints. The problem is a continuous control task with high-dimensional state and action spaces, strict safety requirements, and non-stationary dynamics.

Gao (2014) reported initial results from deploying neural network predictions to guide data center operations. The system used over 100 sensor readings, including temperatures at multiple locations, coolant flow rates, power draw, and external weather conditions, to predict Power Usage Effectiveness (PUE).[4] Operators used these predictions to adjust equipment setpoints. This assisted approach achieved a 40% reduction in cooling energy and a 15% improvement in overall PUE.

Lazic et al. (2018) subsequently deployed a fully autonomous system using model-predictive control with learned dynamics. The state space consists of temperatures, flow rates, and equipment states across multiple zones. Actions are setpoints for chillers, fans, and pumps. The system learns a neural network dynamics model $\hat{f}$ from operational data:

$$s_{t+1} = \hat{f}(s_t, a_t) + \epsilon_t. \tag{3}$$

At each control interval (5 minutes), the system optimizes over a horizon of future states using the learned model, subject to temperature constraints:

$$\min_{a_0,\ldots,a_H} \sum_{h=0}^{H} c_{\text{energy}}(a_h) \quad \text{s.t.} \quad T_{\min} \leq T_{\text{zone},i}(\hat{s}_h) \leq T_{\max} \; \forall i, h. \tag{4}$$

[4]PUE is the ratio of total facility energy to IT equipment energy. A PUE of 1.0 means all energy goes to computing; the industry average is approximately 1.6.

The MPC approach provides interpretability and safety guarantees that pure model-free RL cannot. Actions are projected onto the feasible set defined by temperature constraints, with uncertainty estimates from the learned model used to maintain conservative margins. Human operators can override at any time. Table 2 summarizes the reported performance.

Table 2: Google data center cooling results from Gao (2014) and Lazic et al. (2018).

Metric	Improvement	Method
Cooling energy	-40%	Neural network predictions
PUE	-15%	Assisted operation
Additional efficiency	-12%	Autonomous MPC
Temperature violations	0	Safety constraints

The system operates continuously across multiple data centers, processing millions of sensor readings per day and controlling thousands of setpoints. The learned model adapts online to seasonal variations, equipment degradation, and workload changes. The lesson is that model-based RL with explicit safety constraints is better suited to critical infrastructure than model-free approaches that require extensive exploration.

0.3 Hotel Revenue Management

Hotels face a dynamic pricing problem with perishable inventory, since rooms unsold by check-in yield zero revenue. Classical revenue management uses booking limit controls derived from dynamic programming under parametric demand assumptions. RL offers an alternative that learns pricing policies from data without specifying demand functional forms.

The problem is a finite-horizon MDP. The state (t, I, d) consists of days until check-in, remaining rooms, and a demand intensity indicator. Actions are discrete price levels. Booking arrivals follow a Poisson process with intensity depending on demand conditions; each arrival books with probability decreasing in price. The objective is to maximize total revenue:

$$V_t(I, d) = \max_{p \in \mathcal{A}} \mathbb{E}\left[p \cdot B + V_{t-1}(I - B, d') \mid I, d, p\right], \tag{5}$$

where B is the number of bookings at price p and d' is the next-period demand state.

Gallego and van Ryzin (1994) established the analytical benchmark for dynamic pricing with Poisson demand. Under their intensity control framework, the optimal price

trajectory can be characterized when demand follows specific functional forms. The Gallego-van Ryzin solution provides a theoretical upper bound, and any practical pricing system should approach this performance in settings where the model assumptions hold.

Chen et al. (2023) applied deep Q-learning to hotel revenue management, training on simulated booking data under parameterized demand models. The DQN agent uses state features (normalized time, normalized inventory, demand indicators) and outputs Q-values for each price level. Table 3 compares performance against baselines on held-out test periods.

Table 3: Hotel revenue management results from Chen et al. (2023).

Method	Mean Revenue	vs. Fixed Price
Fixed price	2,847	–
Myopic pricing	2,962	+4.0%
EMSR-b heuristic	3,156	+10.9%
Exact DP	3,241	+13.8%
DQN	3,198	+12.3%

The DQN agent achieves 98.7% of the DP optimal revenue while scaling to capacities where exact DP is infeasible.[5] Training time is nearly constant in hotel capacity, whereas DP computation grows linearly. The learned policies exhibit interpretable patterns, with prices increasing as inventory depletes and as check-in approaches, with larger adjustments during high-demand periods.

0.4 Financial Order Execution

Optimal execution of large equity orders requires balancing execution risk (adverse price movements while holding inventory) against market impact (price deterioration from trading aggressively). The theoretical benchmark is the Almgren-Chriss framework (Almgren and Chriss, 2001), which derives optimal deterministic schedules under linear impact assumptions. Real markets exhibit state-dependent liquidity that these assumptions miss.

Nevmyvaka et al. (2006) applied tabular Q-learning to execution on real limit order book data from NASDAQ stocks (AMZN, QCOM, NVDA). The state consists of time re-

[5] EMSR-b (Expected Marginal Seat Revenue, variant b) is the standard heuristic in airline and hotel revenue management. It computes booking limits by comparing the expected marginal revenue of reserving a unit for a higher fare class against selling it now.

maining, inventory remaining, bid-ask spread, and signed volume imbalance.[6] Actions are trade sizes or the decision to wait. The reward is the negative implementation shortfall,[7] the difference between execution price and a benchmark. The agent learns to condition on market microstructure signals, trading aggressively when spreads are narrow, wait when order flow predicts favorable price movement, accelerate near the deadline.

Table 4: Execution results from Nevmyvaka et al. (2006) on NASDAQ stocks. TWAP is the time-weighted average price baseline (equal-sized trades at uniform intervals).

Stock	RL vs. TWAP	RL vs. Almgren-Chriss	Data
AMZN	−18.2%	−14.6%	500 days LOB
QCOM	−22.1%	−19.3%	500 days LOB
NVDA	−15.8%	−12.1%	500 days LOB

The negative percentages indicate lower costs for the RL agent. Improvements of 12–19% over the Almgren-Chriss baseline are economically significant. For institutional traders executing billions of dollars annually, this represents substantial savings. Several investment banks have subsequently deployed production execution algorithms built on these ideas, using deep RL with richer state representations.

0.5 Supply Chain Inventory Management

Multi-echelon inventory systems coordinate ordering decisions across supply chain stages, where each stage's order becomes the next stage's incoming shipment. A retailer orders from a warehouse, which orders from a distribution center, which orders from a factory. The sequential nature creates complex dependencies, since an order placed upstream today affects downstream availability many periods later. The state space grows exponentially in the number of echelons, with K stages each having M possible inventory levels yielding M^K states. Classical inventory theory provides elegant solutions for special cases, most notably the echelon base-stock policy of Clark and Scarf (1960).

The MDP formulation for a serial K-echelon system is as follows. The state $s = (I_1, \ldots, I_K)$ records on-hand inventory at each stage, where stage 1 faces customer demand

[6] The limit order book records all outstanding buy and sell orders at each price level. Signed volume imbalance is the difference between buy and sell order volume near the best prices, normalized by total volume; positive imbalance typically predicts short-term price increases.

[7] Implementation shortfall measures the total cost of executing a trade relative to a benchmark price, typically the mid-quote at order arrival. It captures market impact, timing cost, and opportunity cost of unexecuted shares.

and stage K is the most upstream. The action $q \in \{0, 1, \ldots, Q_{\max}\}$ is the order quantity placed at stage K. Demand $D_t \sim F_D$ arrives at stage 1 each period; unfilled demand is backordered. Shipments flow downstream, with stage k receiving what stage $k+1$ shipped in the previous period. The per-period cost combines holding, backorder, and ordering components:

$$c_t = h \sum_{k=1}^{K} I_k^+ + b \cdot (D_t - I_1)^+ + c_o \cdot q_t, \tag{6}$$

where $I_k^+ = \max(0, I_k)$ is on-hand inventory, $(x)^+ = \max(0, x)$, h is holding cost per unit, b is backorder cost per unit, and c_o is ordering cost. The objective is to minimize expected discounted total cost.

Gijsbrechts et al. (2022) conducted a systematic evaluation of deep RL against classical base-stock policies across lost sales, dual sourcing, and multi-echelon configurations. The classical benchmark is the echelon base-stock policy, in which each stage k maintains an echelon inventory position and orders to bring this position to a target level S_k.[8] For single-echelon systems with backorders, the optimal base-stock level is given by the newsvendor critical fractile $S^* = F_D^{-1}(b/(b+h))$. Table 5 summarizes results from their multi-echelon experiments.

Table 5: Multi-echelon inventory results from Gijsbrechts et al. (2022).

Echelons	DRL Cost Gap vs. Base-Stock	States	Convergence
2	+1.2%	$\sim 10^2$	Achieved
3	+6.8%	$\sim 10^3$	Achieved
4	+12.3%	$\sim 10^5$	Partial
6	—	$\sim 10^8$	Failed

The results are sobering for RL proponents. In settings where classical solutions exist and can be computed, RL adds little value and often underperforms. The base-stock policy, derived analytically for idealized settings, remains highly competitive when properly calibrated. RL struggles particularly with multi-echelon coupling, where orders at upstream stages affect costs at downstream stages many periods later, creating difficult credit assignment problems.[9] The authors report that DRL required millions of train-

[8] Formally, the echelon inventory position at stage k equals on-hand inventory at k plus all inventory at downstream stages $1, \ldots, k-1$ plus in-transit inventory, minus backorders. The echelon base-stock policy of Clark and Scarf (1960) is optimal for serial systems with linear costs and backorders.

[9] Credit assignment refers to the difficulty of determining which past actions caused a delayed reward or cost.

ing transitions to approach performance levels that base-stock policies achieve through closed-form calculation. The value proposition for RL in inventory management lies in problems where analytical solutions are unavailable, such as non-stationary demand, complex operational constraints, or settings where the cost structure does not admit tractable decomposition.

0.6 Real-Time Bidding

Real-time bidding for display advertising presents a budget pacing problem in which an advertiser must allocate a fixed budget across a sequence of auctions to maximize conversions. Wu et al. (2018) formalized this as an MDP where the state includes remaining budget, time elapsed, and recent win rates, with actions being bid multipliers. The RL agent learns to adapt bidding intensity to market conditions, increasing bids when competition is low and conserving budget when prices spike. Simulation experiments calibrated to real RTB data showed improvements over rule-based pacing algorithms.

0.7 Simulation Study: Bus Engine Replacement

I conclude with a simulation demonstrating that RL matches dynamic programming on a classical economics benchmark. The bus engine replacement problem, introduced by Rust (1987), models a fleet manager's monthly decision whether to replace engines based on accumulated mileage. Replacement incurs a fixed cost but resets mileage to zero; continued operation incurs maintenance costs increasing in mileage.

I extend the single-engine problem to a fleet of N engines with a capacity constraint limiting replacements per period. The state $s = (m_1, \ldots, m_N)$ records discretized mileage for each engine. Actions are subsets of engines to replace, subject to the capacity constraint. The per-period cost is $c(s, a) = \alpha \sum_i m_i + \beta|a|$, where α is the operating cost per unit mileage and β is the replacement cost.[10] Mileage evolves deterministically; replaced engines reset to $m = 0$; others increment by one bin. With $M = 6$ mileage bins, the state space is 6^N, yielding 1,296 states at $N = 4$, 7,776 at $N = 5$, 46,656 at $N = 6$. Value iteration is feasible for $N \leq 5$.

In multi-echelon systems, an upstream ordering decision today may not affect customer-facing costs for many periods, making it difficult for RL to learn the causal connection.

[10]The mileage-dependent operating cost $\alpha \sum_i m_i$ follows Rust's original specification $c(x, \theta_1) = \theta_{11} x$, creating a non-trivial threshold replacement policy. The fleet extension uses deterministic mileage increments to isolate the combinatorial scaling challenge that arises from the joint state of multiple engines.

Table 6 compares dynamic programming, DQN, and heuristic baselines across fleet sizes.

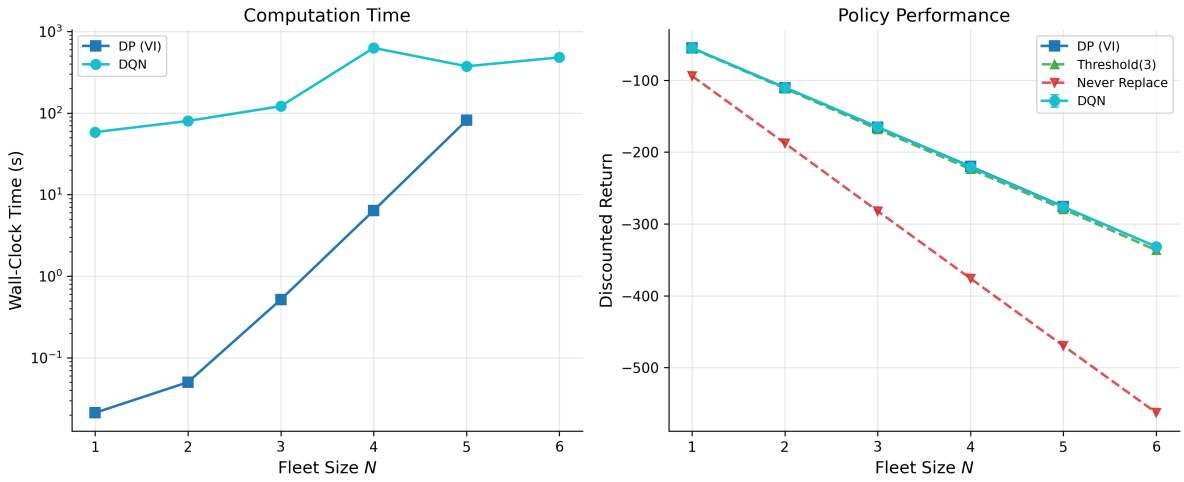

Figure 1: Bus engine replacement benchmark. Left: computation time vs. fleet size (log scale). Right: discounted return vs. fleet size for DP, DQN, and heuristic baselines.

Table 6: Bus engine replacement results across fleet sizes.

N	$\lvert\mathcal{S}\rvert$	DP Time	DP Return	DQN Return	Thresh(3)	Never Replace
1	6	0.02s	-55.1	-55.1 ± 0.1	-55.7	-94.0
2	36	0.05s	-110.3	-110.0 ± 0.3	-111.4	-187.7
3	216	0.52s	-165.3	-165.0 ± 0.1	-168.3	-282.1
4	1,296	6.38s	-220.0	-220.8 ± 0.1	-223.8	-376.0
5	7,776	81.33s	-275.7	-276.2 ± 0.1	-279.5	-469.7
6	46,656	—	—	-331.7 ± 1.1	-336.6	-562.8

For $N = 1$ through 5 where both methods are computable, DQN matches DP within 1% of the optimal discounted return. At $N = 6$ (46,656 states), DP is infeasible but DQN produces a policy. The threshold heuristic, which replaces engines above a mileage cutoff, provides a reasonable baseline but cannot account for capacity constraints or the joint state of multiple engines. The never-replace heuristic performs poorly due to accumulated mileage costs, confirming that the cost structure creates a non-trivial replacement decision.

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